

More Than Just Limits: How Technology Can Support Parents in Regulating Children's Eating Behaviors at Family Mealtimes

Yang Chen

sonnechen95@gmail.com

National University of Singapore

Singapore 

Ching Chiuan Yen

didyc@nus.edu.sg

Division of Industrial Design, Keio-NUS CUTE Center,

National University of Singapore

Singapore 

ABSTRACT

To alleviate concerns regarding technology use at family mealtimes, this study explored how specific technological probes (including augmented utensils, screen-based designs, and smart objects) can aid parents in regulating children's eating behavior. Our two-phase investigation included remote observations of 25 families with children aged 3-7 to unobtrusively understand parent-child interactions. This was complemented by retrospective semi-structured interviews to delve into food parenting practices and technology-related preferences and concerns. Our study uncovered a mismatch between parents' perceptions and the actual eating behaviors of children, emphasizing the significance of technology in both regulating eating habits and offering objective insights to parents. While screen-based technologies were identified as the most feasible and desirable option for managing children's eating, we proposed enhancing their functionality by integrating parental involvement and control mechanisms along with mitigating adoption barriers (e.g., technology dependence and distraction) to better support parental intervention during family meals.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**; **Empirical studies in ubiquitous and mobile computing**.

KEYWORDS

eating intervention, children, family mealtime, technology-mediated intervention

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1 INTRODUCTION

The preschool years are a critical time for establishing healthy eating habits, which, once formed, often continue into adulthood

[2]. However a range of eating behaviors, ranging from picky eating and food neophobia [22], emotional over-and under-eating [1, 5, 12], to common mealtime misbehaviors [11] (e.g., food throwing, or not staying seated at the table), are prevalent. Such behaviors often lead to more authoritative parental approaches in behavior management, which can lead to tension between the two parties [42]. Consequently, this adversely affected children's physical and mental development [4, 35] as well as the overall atmosphere of family meals [16].

Technology-mediated interventions, incorporating smart sensors and multisensory feedback, have shown promise in engaging children in healthier eating practices. For instance, *EducaTableware* [26] is an interactive tableware that produces auditory feedback to enhance the enjoyment of eating and promote certain behaviors during mealtime. Similarly, *Cat Fork* [9] featured a cat-shaped fork that interacted with a plate, lighting up in a cat face pattern to motivate children to use utensils during meals. However, the integration of technology into mealtime is not without its challenges. A prevalent concern exists among parents that the use of technology (e.g., mobile phone) could lead to unwanted distraction, thus diminishing meaningful family interaction and disrupting the mealtime sanctity [13, 18, 41]. This concern is echoed in expert recommendations advocating for media-free spaces during meals to mitigate risks such as obesity and antisocial behavior [13, 18, 24].

There has been a growing interest in HCI in exploring how parents perceive technology use within the family, both generally and in specific contexts such as mealtimes, to uncover ways to alleviate initial reluctance and find potential acceptance of technology within a family environment. Studies have shown that parents' approach to technology is not about outright prohibition but about establishing specific guidelines that balance its benefits with the need for interpersonal connection in family interactions [18, 20, 29]. These attitudes and decisions towards technology adoption are dynamic and influenced by various personal (e.g., emotions) and situational factors (e.g., immediate needs) [28, 31]. Recent research further emphasized this diversity [10], suggesting that reluctance doesn't always translate to a total prohibition. Rather, this hesitancy signaled a call for better-designed technological solutions that can alleviate parents' concerns such as potential intrusions and distractions. Inspired by this evidence, this study aims to **investigate which specific technologies might be more favorably received and could augment existing food parenting strategies, thereby aiding parents in tackling particular eating issues among children**.

Our study involved remote observation of 25 families with children aged 3-7 during mealtime, complemented by semi-structured

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retrospective interviews. This dual-method approach provided an in-depth understanding of parent-child interactions during family mealtime. We identified a notable discrepancy between parents' perceptions and the actual eating challenges of their children, particularly the underexplored issue of slow eating. Parental approaches to managing their children's varied eating issues were systematically classified within a spectrum of parental control and involvement (see Fig 1). This adapted from Vaughn et al.'s framework [43], which was initially designed for broad food parenting practices, to our specific context of eating behavior regulation. Interestingly, our findings showed a continuum of parental control in children's eating regulation reflected in the strategic use of screen-based technology during family mealtimes. This ranged from supportive approaches (e.g., entertainment), to structured educational tools, and even to coercive control (e.g., complete prohibition). This versatility positioned screen-based technology as highly appealing for assisting parents in regulating children's eating behaviors, as compared to other technological options like augmented utensils and smart objects (detailed in 2.2). Furthermore, we discussed how screen-based technology could be further refined to overcome potential adoption barriers in family mealtimes. We recommended a shift from targeting instrumental benefits for children solely in technology-mediated interventions to a more holistic approach that incorporated family dynamics. This involved a significant emphasis on parental involvement, aiming to bolster parent-child-technology collaboration in cultivating healthy eating behavior. Furthermore, we underscored the necessity of integrating control mechanisms tailored for both parents and children. Such a strategy should promote self-regulation in children, utilizing supportive and educational elements, thereby effectively mitigating the likelihood of technology dependence.

Unlike research on parents' perspectives about general technology use during family mealtimes [10], our work contributes to a nuanced view of how specific technologies can augment parental strategies for managing children's eating behaviors. This understanding identified a novel perspective that goes beyond traditional parent-child or technology-child-focused methods to emphasize the synergistic effects of parental involvement, technological support, and child engagement for eating intervention during family mealtimes.

2 METHOD

We conducted remote observations with 25 families during one of their mealtimes, followed by a semi-structured retrospective interview with the same parents involved. The purpose of observation is to obtain an objective understanding of the dynamics of family meals, involving at least one parent and their children, to accurately represent their typical family dining experiences. Following this, we engaged parents in the interviews, where we presented the video recordings of their family mealtimes along with observational notes highlighting their specific behaviors and interactions. This approach enriched our understanding of the underlying reasons for parents' specific food practices in addressing specific eating challenges. Additionally, it provided insights into how technology could be leveraged effectively in this context.

2.1 Participant

We recruited parents with children aged 3 to 7, a critical period for establishing eating habits [3]. As parents play a vital role in shaping their children's eating habits during these formative years, this age range presents an ideal window for impactful habit formation and eating intervention [2]. No specific exclusion criteria were used, except for children with eating disorders (e.g., anorexia nervosa), since these conditions could result in parents' nudges not generally applicable to healthy eating. In the end, 25 parents from different families participated in our study, with the majority being mothers ($N=18$, 72%). Twelve (48%) of these participants had a single child, and two (8%) had three children, which resulted in our study including 24 boys and 16 girls with an average age of 5.7 years (Median=5, $SD=1.84$).

2.2 Data collection

Phase 1: Remote Observation. For a realistic capture of family mealtime dynamics, we directed parents to position their phones at a designated angle within their homes. Throughout this phase, not only did the system (e.g., Zoom) record the entire meal, but we also took screenshots of specific parent-child interactions, which were then annotated to document key moments with descriptions. These annotated photo documentation (see Appendix B as examples) [19] provided an additional layer of observation, eliciting rich parents' coping strategies in response to their children's challenging eating habits.

Phase 2: Semi-structured Retrospective Interview. We structured the interview into two main sections, see Appendix A for interview questions. The first section, complemented by video recordings and observational notes, concentrated on examining parents' food parenting practices, the perceived effectiveness of these strategies, and their response to situations where children did not follow their eating guidelines. In the second section, we shifted focus to parents' views on using technology to support the regulation of their children's eating behaviors. For parents to gain a better understanding of how technology-enabled interventions can be implemented, we divided existing HCI studies introducing various strategies for promoting healthy eating among children into three groups: 1) Augmented utensils [15, 46], which incorporate technologies like projectors in eating tools for interactive prompts; 2) Screen-based designs [17, 25], using screen technology to display prompts and mimic children's eating behaviors and 3) Smart objects [33, 37], introducing artifacts like robots as eating companions. Voice interfaces were excluded due to their lower preference in family dining settings [10]. We selected two examples from each category, demonstrating positive impacts on children's eating behavior regulation from the HCI community (see Appendix C). Parents were asked about their preferences for particular technologies, their feasibility for home integration, and their thoughts on potential improvements. The entire study spanned between 42 to 71 minutes. Each participating parent received a 20 SGD gift card.

2.3 Data analysis

Upon consolidating 25 transcribed interview notes with observational notes detailing parent-child mealtime interaction, we performed an inductive thematic analysis [7]. Initially, the first three

interview transcripts were open-coded by two researchers independently using the qualitative data analysis software ATLAS.ti [45]. This process resulted in preliminary codes such as "eating together with the television off" and "causal conversation during mealtime" which then subsequently clustered into potential code groups such as "family mealtime routine". Descriptive theme memos were also created, providing a comprehensive understanding of each code group and being grounded in the corresponding transcript excerpts. To ensure the reliability of coding, two additional interview transcripts were independently analyzed by authors, enabling the incorporation of any new emerging themes and necessary updates to theme memos. Through multiple rounds of iterative coding, review sessions, and discussions, the themes were refined and modified until a consensus was reached among the authors: for instance, parents' coping strategies (i.e., verbal instruction with consequences and benefits, utilization of incentives). The first author independently coded the remaining 20 transcripts, resulting in 190 codes that converged on four main findings, which are discussed below.

3 RESULTS

This section discussed the key themes that emerged from our analysis, highlighting the dynamic nature of family mealtimes. It particularly focuses on parents' coping strategies and perspectives on technological aids for regulating children's eating behavior. To provide quotes from participants, we refer "P" for parents followed by participant numbers.

3.1 Children's Eating Issues: the Potential Contributor and its Dynamic Nature

3.1.1 Discrepancies in Parental Perception vs. Reality of Children's Eating Patterns. In most cases, slow eating emerged as a prevalent concern not only reported by parents (16 out of 25) but also observed based on our observation. It typically manifests as prolonged chewing, food pocketing (the act of retaining food in the mouth without swallowing), and substantial pauses between bites. These patterns are in line with a recent review paper [8] examining eating rate correlates. It is important to note that parents' subjective assessment of a child's eating speed, often based on comparisons to their own pace or that of other children, can lead to considerable inaccuracy in determining what the true situation is. For instance, P5 expressed concerns about her children's slow eating; however, our observations indicated that their eating pace was normal, even slower than other observed children within a single meal. The discrepancy between our observational findings and parents' reports suggests that parents' perceptions of undesirable eating behaviors may differ from the issues that actually exist. Contrastingly, picky eating, identified as a concern by 13 out of 25 parents, presents a more clear-cut problem. This behavior is characterized by reluctance towards certain foods, possibly due to their taste and palatability, and often coexists with other challenging behaviors like food refusal and slow eating. Additionally, behavioral challenges at the table, such as difficulty staying seated, are more observable and consistent.

3.1.2 The Interplay of Location, Social Dynamics, and Individual Factors in Shaping Children's Eating Behaviors. Children's eating behaviors are not static but rather dynamic, shaped by a complex

interplay of location, social dynamics, and individual factors. A majority of parents (16 out of 25) reported that their children's eating issues were mitigated when eating in external environments (e.g., school, restaurant, grandparents' house) compared to the home setting. One parent (P8) noted, *"he doesn't eat vegetables... he eats vegetables at school...and at his grandma's house."* The home environment, characterized by familial relationships, may lead to more individualistic eating behavior, where children are characterized as becoming slightly *"more mischievous"* (P8), *"more willing to test her limits"* (P17), and are comfortable expressing their food preferences without feeling obligated to do so. While in external environments, children's eating behaviors are influenced by the presence of peers, authority figures such as teachers, and structured rules surrounding mealtimes that lead them to adapt their eating behaviors in ways that are less challenging than what they would observe at home [32, 36]. Although location and social dynamics influence eating behaviors among children, they are not fully explained. As reported by half of families with more than one child, siblings who share the same home environment and potentially similar social influences can exhibit different eating behaviors. This diversity challenges the idea of the modeling effect, which suggests that individuals tend to emulate the eating behavior of those around them [34]. These differences may be attributed to individual personality traits and preferences, or varied caregiving approaches. As P19 mentioned *"Because my older one. I fed him myself... for my youngest one, who is taken care of mostly by my helper. They let him do whatever he wants."*

3.2 Parental Strategies in Regulating Children's Challenging Eating Habits

3.2.1 Universal Strategies Across Varied Eating Behaviors. While each family adopted its own methods at varying levels of control and involvement to guide and influence their children's eating issues, however, the broader strategies employed by parents were not particularly unique (see figure 1). Instead of enforcing strict rules or manipulation (e.g., forbidden talking, set mealtime limits), most parents (22 out of 25) employed more relaxed, persuasive, and flexible strategies to subtly encourage children to adopt certain eating behaviors. This approach was largely motivated by the parent's intention to help children develop a healthy perception of mealtime, aiming to foster *"engagement in eating"* (P20) rather than *"creating a negative association with food"* (P8). This is achieved through various means: the provision of various incentives, the integration of entertainment elements like video play during meals, and the customization of meals to align with children's dietary preferences. Verbal instruction, though widely used among parents, exhibits a diverse range in both their application and underlying reasons, spanning from straightforward explanations to more intricate forms of storytelling or role modeling. For instance, parents like P7, employ practical explanations to encourage rapid eating, like *"if cool the food taste less nice anymore"*. In contrast, others, like P13, provide more comprehensive, health-focused reasons: *"Carrots are rich in vitamin C, which helps you prevent sneezing and a runny nose."* In certain instances, these verbal cues evolve into a form of storytelling or role modeling. An example is encouraging a child

	No Control	Permissive Control	Structured Control	Coercive Control
	Parents do not exercise any form of control over their child's eating behaviors	Parents who set few rules and tend to be very lenient, taking on more of a friend role than a parent role	Parents set clear expectations and rules for children's behavior	The use of force, manipulation, threats, or punishments in order to control a child's eating behavior
Active Involvement <i>Active participation, monitoring, or control</i>	• Various food options • Vitamins & Supplements • Encouragement and Praise	• Rewards or incentives (e.g., dessert, TV, phone) • Adjust food size/flavor toward children • Verbal instructions: consequences and benefits • 'One-Bite Rule' • Modelling	• Set clear rules and consequences (e.g., children should wash the plate on their own) • Limited food accessibility • Verbal instructions without reasons • Meal and Snack schedule	• Strict rules and restrictions (e.g., forbidden talk) • Set the timing (e.g., meal duration) • Pressure and Force to eat (e.g., scoop and feed) • Directly interrupting (e.g., pause the meal)
Inactive Involvement <i>Creating a supportive environment or offering guidance by setting environment conducive to healthy eating</i>	• A pleasant and relaxed ambience (e.g., reading books, playing music, TV)	• Distraction (e.g., TV) • Engage children in food preparation	• Distraction adjusted based on eating (e.g., turn off the TV if certain unwanted eating behavior occur)	• Avoid the usage of any potential distraction (e.g., TV)

Figure 1: We categorized parental strategies for nudging their children's eating habits into two dimensions: the level of parental involvement (active and inactive) and the degree of control exerted (ranging from non-control to coercive control). Adapted from Vaughn et al. [43]

by drawing parallels with a sibling, using aspirational narratives "eat more can be strong like your sister"(P19).

3.2.2 Screen Time: A Reflective Measure of Parental Control. Parents' attitudes towards screen use during meals interestingly reflected the varying degrees of control they exercise in managing their children's eating patterns. This spectrum of control ranges from a permissive approach, where screens were viewed as tools for entertainment and family bonding, as seen in P8's family who used video calls to engage with distant relatives during meals, to a more authoritative stance where any form of distraction (i.e., screen) was avoided during family meals. This latter approach, adopted by the majority of families (15 of 25) is because of the ubiquitous exposure of digital devices, ranging from traditional televisions to smartphones and tablets in children's daily lives, which has motivated parents to maintain the sanctity of mealtimes. In a more structured control strategy, nearly a third of parents tactically employ digital devices, either as a means of distraction or as an incentive to encourage changes in eating behavior. The allowance of screen time at the meal table is closely tied to the child's behavior, with adjustments being made as needed.

3.3 Parental Perception of the Adoption of Technology-based Intervention

3.3.1 The Reluctance and Acceptance Paradox in Screen-Based Technologies. Screen-based interventions emerged as highly feasible and well-accepted for promoting healthy eating during family meals, with nearly half of the parents (12 out of 25) favoring this approach. The empirical evidence supported the widespread adoption of mobile applications in the field of children's eating interventions. Interestingly, a significant proportion (64%) of those who opted for screen-based solutions initially had reservations about introducing screen time during meals. The engaging and interactive features of these technologies, such as animated characters like a panda eating [25], played a key role in alleviating these concerns. These dynamic elements, offering more than just "passive video watching"(P5), with various visual cues, animal avatars, and flexible storylines, could counter the novelty effect and make the experience more appealing

to children, potentially addressing the limitations of current strategies parents employ. The specific interactions, designed to mimic children's eating behaviors, are particularly valued by parents, as they foster emotional connections and a sense of autonomy, crucial for children at developmental stages like 5 or 6 years old [39]. In terms of practical implementation, these interventions could be easily implemented by parents using their smartphones. By inadvertently setting aside their own phones during meals, parents benefited from greater engagement with their children, as P21 explained, "I had to reduce my use of my phone during mealtime, allowing me to focus more on dining with my children." However, there was concern about device familiarity potentially distracting children from eating and extending meal duration, despite suggestions made by parents to place screens farther away or to use parental controls. These issues led to smart objects being the second most popular option for integrating family mealtime.

3.3.2 Divergent Parental Views on Smart Object Technologies: Balancing Novelty with Parental Roles. Seven out of 25 parents showed a preference for smart objects like robots as a viable alternative, primarily those who were hesitant about using screen-based media during meals. Intriguingly, the attributes considered advantageous by some parents were viewed as drawbacks by others. For example, a group of parents valued the novelty and rarity of robots, believing that these characteristics would sustain maintain the novelty effect, keeping children engaged and less likely to "become bored", as opposed to mobile devices, which children actually "see it quite often"(P7). Other parents, however, expressed concern that this very novelty may backfire. In their view, the rare appearance of robots might overly intrigue children, causing them to become overly preoccupied with the robot, thereby diverting their attention from eating.

Similarly, while some parents appreciate the potential of smart objects to provide interactive, human-like engagement, particularly when they are preoccupied, the concept of robots as "substitutes for helpers or siblings" raises concerns about technology replacing parental roles. P4 highlighted this worry, stating, "I don't want my kid to rely on a robot for eating, nor do I want to be replaced by a robot."

This tension emphasizes a vital insight: the importance of ensuring that technological interventions in family settings supplement rather than replace parental involvement and interaction. The direct interactions between parents and their children are unique and irreplaceable, and parents value these interactions as being more diverse and meaningful than those with pre-programmed robots. Smart objects are limited in a number of ways, including their *"less visually appealing"* (P6) nature, inability to participate in *"dynamic conversations"* (P16), and predominantly fixed functionality that *"does not allow for customization"* (P21). These parental observations further reinforced the importance of technology augmenting and enhancing, rather than substituting for, parental responsibility.

3.3.3 Preference for Simplicity Over Complexity in Augmented Utensil. Augmented utensils are considered the least desirable technologies for promoting healthy eating among children. Although 6 out of 25 parents viewed these technologies as a means of making eating more enjoyable, engaging, and effective, P17 noted, *"kids will not ignore it because they must look at the plate to eat."* The simplicity of this approach was a key factor. This type of invention, however, is based on simple mechanisms, such as children eating to uncover new images, facilitating story development, or winning a digital budget, which is already implemented by parents in their existing strategies in a simpler, no-technical manner, making this approach less desirable. As P2 mentioned, *"some plates now already have a similar idea like if you eat until the bottom, you can see some designs"*. Additionally, practical concerns were raised regarding the incorporation of augmented utensils into the family mealtime. For instance, projector-based designs might be ineffective when obscured by food, and the use of visual or verbal prompts could inadvertently encourage children to play with their food rather than eat it, further reducing the appeal of augmented utensils.

4 DISCUSSION

Tailoring Technology to Tackle Subjectivity in Children's Eating Speeds During Family Meals

Our empirical findings highlighted the intricate nature of children's challenging eating behaviors, influenced by a blend of factors including the eating location (home vs. school), social dynamics (e.g., the presence of peers and authority figures), and individual characteristics (e.g., personality traits and relationships with caregiver). Two primary behaviors identified by parents were slow eating and picky eating, both tend to be more prevalent at home due to the comfort and familiarity of the environment, leading to a need for tailored home-based intervention solutions.

Picky eating, characterized by obvious signs of food refusal or a limited dietary range, is a well-documented and explored issue [40]. Its clarity and observability have spurred a wealth of research in HCI utilizing multisensory feedback and gamification to broaden children's dietary preferences and foster healthy eating habits during early childhood. For example, Azusa Kadamura et al. [25] developed an interactive fork and mobile app to encourage children's healthy eating habits through storytelling. YJoi Yeong Rae et al. [23] introduced a smart tableware system to motivate children to eat more vegetables willingly. In contrast, slow eating presents a unique challenge due to the absence of standardized benchmarks defining

'fast' or 'slow', rendering it a more subjective concept [8]. Our findings indicated that this subjectivity, often influenced by parents' own eating habits, can lead to inaccurate perceptions of a child's eating speed. This discrepancy poses an unexplored avenue in HCI: **aiding parents in objectively understanding their children's eating pace and promoting an optimal eating rate.** Note that, research has shown that faster eating rates are linked to higher energy intake within meals and correlate with increased body weight and adiposity in children [14]. Therefore, the objective should go beyond simply accelerating a child's eating rate but to establish and maintain a healthy eating pace aligned with the child's natural body rhythm. Also, due to the lack of standardized benchmarks for eating pace, which are influenced by food texture, technology could assist parents in fostering healthier eating habits by providing tailored nutrition and food recognition advice based on the foods children consume.

Redefining Technology's Role in Family Mealtimes: Balancing Practical Solutions with Experiential Benefits

Contrary to the widespread belief that parents limit screen use (like watching videos) to uphold mealtime sanctity, our analysis painted a more complex picture. Screens, in this context, go beyond being mere distractions; they are thoughtfully integrated into family meal routines. They function as tools in food parenting practices, mediums to foster conversation, enablers of digital communication with distant family members, and educational resources. This strategic integration interestingly reflects the varying levels of control parents exercise in their food parenting practices, positioning screen-based designs as versatile tools in regulating children's eating behaviors. Moreover, the ease of setup and the rich interactive nature of screen-based designs have made them particularly appealing to parents, emerging as the most desirable technology form for family mealtimes, which accordance with [10]. In contrast, technologies like augmented utensils and robots, are often criticized for incurring additional costs and requiring ongoing maintenance. **This inclination towards practical, easy-to-adopt, and cost-effective technological innovations** is not only prevalent in general family mealtime settings but is also reflected in our empirical findings as technology-mediated food parenting practices.

While our focus is on the instrumental aspects of technology as a means to control children's problematic eating behaviors during family meals, our findings revealed that screen-based technology, notably promising in food parenting, transcends its initial functional role to provide experiential and emotional benefits, adeptly catering to diverse eating contexts and meeting parental needs. This aligns with the growing field of digital and computational commensality in HCI [6, 30], including a range of innovations from co-dining systems [44] to virtual eating companions [38]. Mobile mealtime assistants, for example, like MAMAS [21], have further demonstrated the power of simple technologies, like mobile phones, in positively transforming family dining experiences. Inspired by these developments, **we recognize the potential of integrating screen-based technology into family meals to serve dual purposes: it provides practical solutions for specific children's**

eating behaviors while simultaneously fostering a more engaging and bonding family dining experience. Using [25] as an example, rather than featuring a panda avatar to mimic and encourage children's eating behaviors, it could be better suited to include a panda family to accomplish both the original goal of the project and the family mealtime context holistically by incorporating parental involvement. This approach suggests that while technology initially targets children's needs, it should also engage parents actively in the system, transforming them from mere observers to facilitators. Thus, technology acts as a bridge, not only addressing children's eating habits from a functional standpoint but also fostering emotional connections within the family, creating a more holistic and inclusive mealtime experience.

Despite the promise of integrating screens into family mealtimes, it is crucial to maintain a balance between engagement and distraction. Effective technology-mediated interventions should captivate children's interest and promote adherence to healthy eating habits, yet they should not dominate or detract from the mealtime experience [27]. For example, gamification strategies, such as using eating as a game controller, are innovative but need to be thoughtfully designed to emphasize healthy eating rather than just gameplay. **The integration of control mechanisms within these technologies is pivotal.** They should empower parents to effectively guide their children's eating behaviors, thereby reinforcing the human elements of parenting and family connection. Concurrently, the technology should offer children a degree of autonomy, promoting self-regulation over simple obedience to the technology. Moreover, these systems should do more than just manage children's eating behaviors; they should also educate and encourage children to appreciate the significance of mealtime and to be mindful of their bodily cues. This necessitates **intervention techniques that are supportive, encouraging, and educational, avoiding any judgmental undertones.** Ideally, these interventions should complement and enhance parents' varied persuasive strategies, incorporating both fun and educational elements. Implementing such an approach would integrate technology into the mealtime experience in a way that is both informative and enjoyable, thereby nurturing a positive relationship with food and eating habits.

5 CONCLUSION

In this work, we present a two-phase investigation involving observations and retrospective interviews to explore the design space of how technology can support parents in managing their children's eating challenges during family meals. We systematically categorized existing food parenting practices and observed how parental control levels influence the strategic adoption and perception of technology in family mealtime contexts. Our findings identified screen-based technology as the most viable and appealing option for integration into current parenting practices, owing to its practicality, cost-effectiveness, and user-friendliness compared to alternative technological probes (i.e., augmented utensils and smart objects). To mitigate concerns regarding potential distractions and technology dependence, we proposed a holistic approach to technology-mediated interventions, by fostering collaborative efforts between parents, children, and technology. We recommended the integration of control mechanisms in the intervention, to not

only reflect family dynamics but promote self-regulation in children.

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A INTERVIEW QUESTIONS

Phase1:

Introduction to Video and Photo Documentation. We have recorded and taken screenshots of specific interactions during your family's mealtime. I'll be showing you these to help us discuss your mealtime practices in detail.

Discussing Specific Behaviors and Reasons.

- (1) In this video/photo, we observed [describe the specific behavior]. Can you tell us more about what was happening here?
- (2) What prompted you to react in this way during this moment?
- (3) Are there any specific reasons or experiences that have shaped this approach to your child's eating [habits/issues]?

Observations on Children's Eating Problems.

- (1) Based on what we observed in the video/photo, how would you describe your child's eating behavior?
- (2) Have you noticed any recurring eating challenges with your child? Can you provide some examples?
- (3) And In what ways do you notice your children's eating problems?

Strategies and Effectiveness for Regulating Eating Behavior.

- (1) In the event that you observe certain eating disorders, what do you do? What strategies do you typically use to address your child's eating behavior as seen in the video/photo?
- (2) What's the timing of your intervention? Will you intervene immediately once you observe certain behavior, or will you do it later afterward?
- (3) Will you tell your child the reason behind your intervention and the potential consequences certain behavior could bring? Why?
- (4) Is there a reason why you choose certain ways to regulate your child's eating issues? Is it going to work? Depending on the intervention, how will your child react?
- (5) What challenges do you face when you intervene in your children's problematic/ unwanted eating behavior (i.e., immediately or afterward)? How will you react?
- (6) What makes a good prompt? Are there any particular approaches or tactics that you've found to be more successful than others? How effective do you think your prompt is? How do you know that?

Phase2:

Technology in Mealtime:

- (1) There's growing interest in using technology to promote healthy eating habits in children. Considering the examples I've provided, what are your thoughts on integrating technology into family mealtimes?
- (2) What types of technology do you think could assist you? And Why?
- (3) Do you consider using [certain technology] in your family mealtime? Why? What parts of the technology should be improved if it is going to be used during your family mealtime? Why?

B PHOTO DOCUMENTATION EXAMPLES

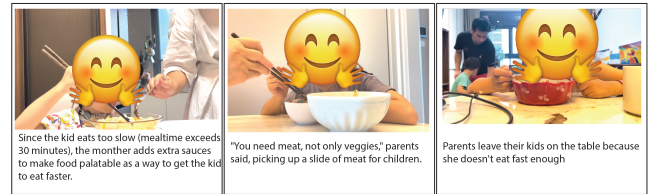


Figure 2: Notes taken by researchers during remote observation of family mealtimes

C EXAMPLES OF TECHNOLOGY-MEDIATED EATING AMONG CHILDREN

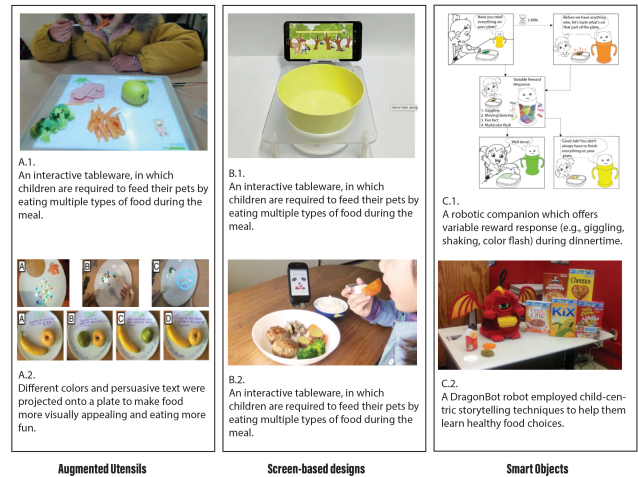


Figure 3: Examples: 1) Augmented utensils: A.1. [46] and A.2.[15], which incorporate technologies like projectors in eating tools for interactive prompts; 2) Screen-based designs: B.1.[17] and B.2.[25], using screen technology to display prompts and mimic children's eating behaviors and 3) Smart objects: C.1. [37] and C.2.[33], introducing artifacts like robots as eating companions.